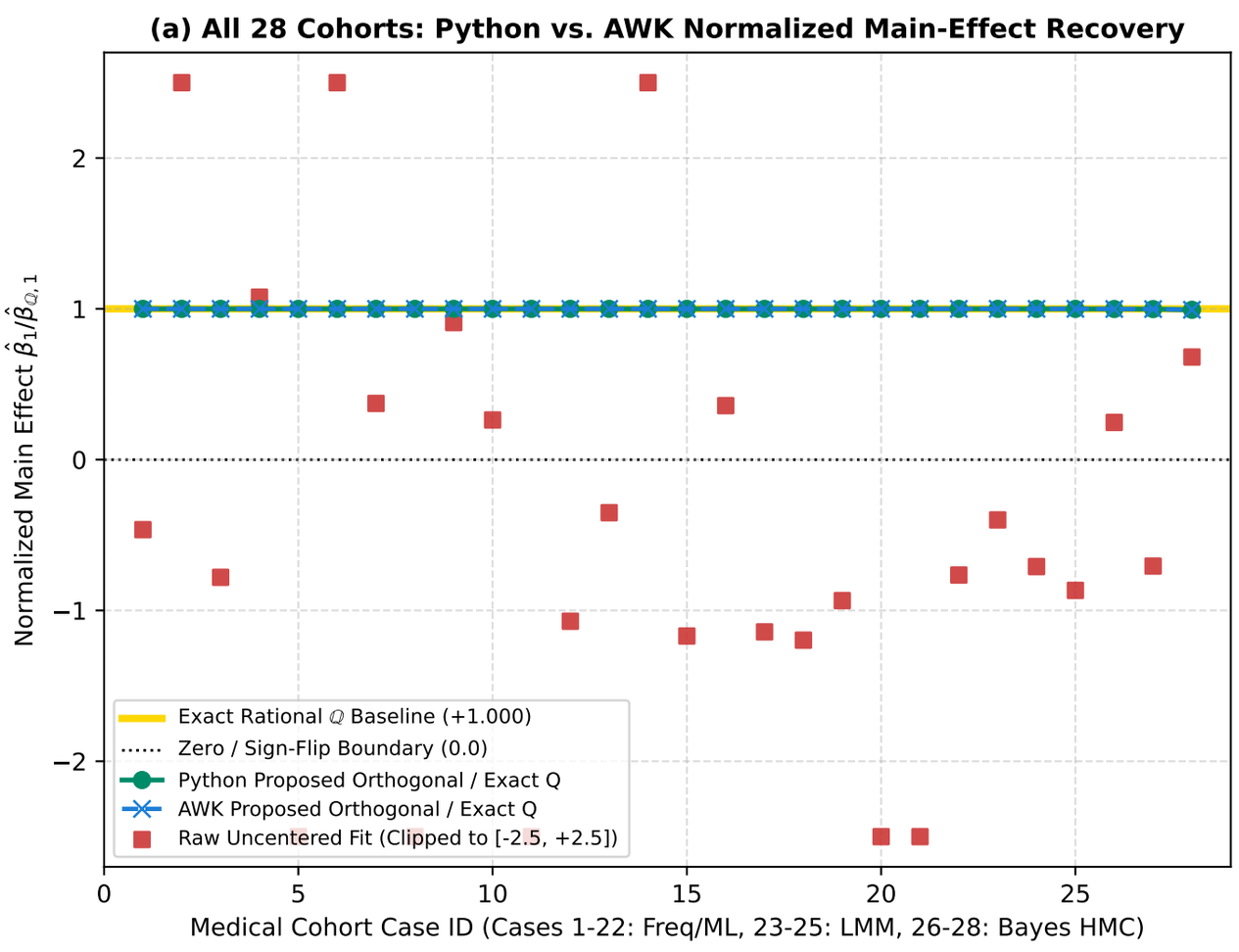
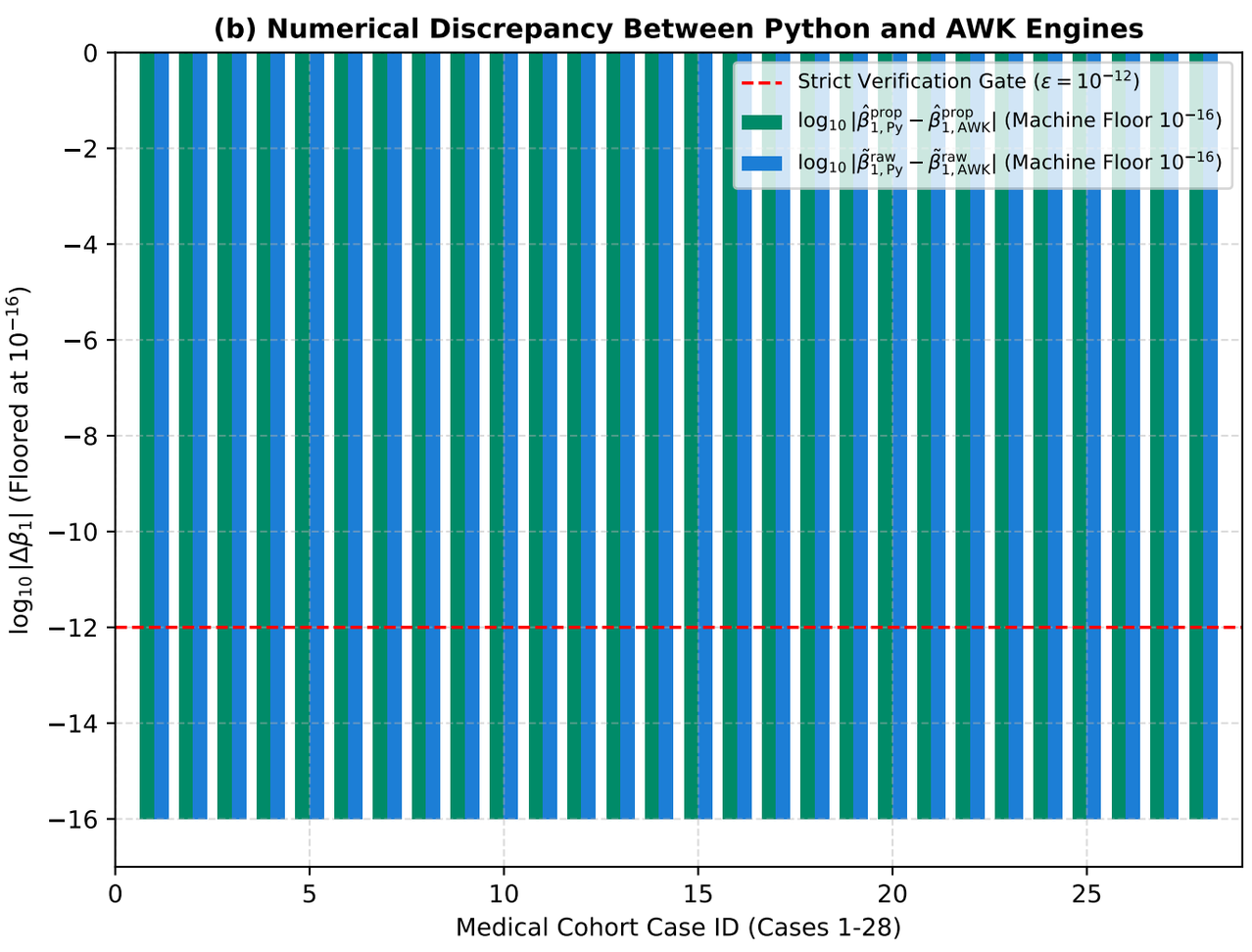
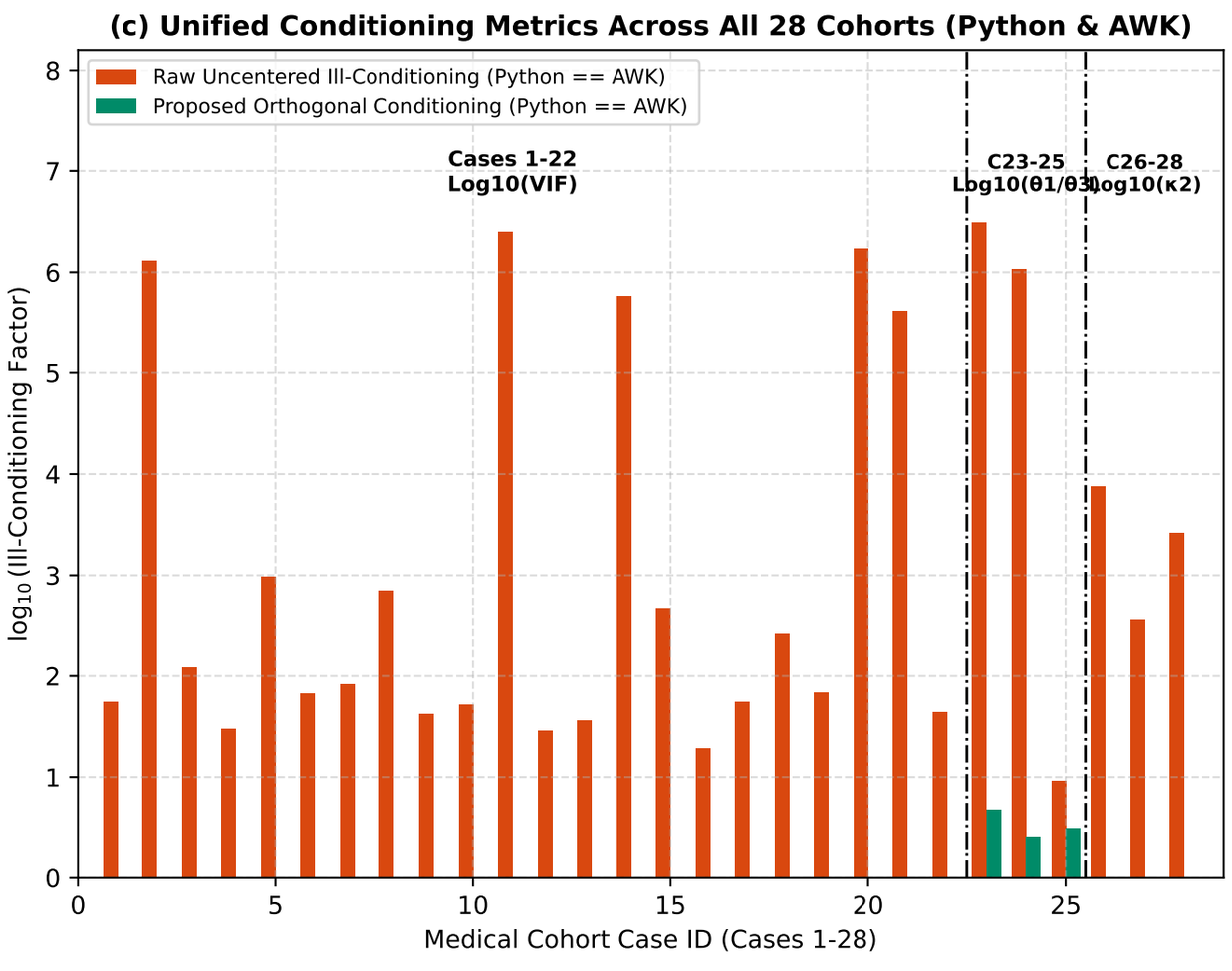
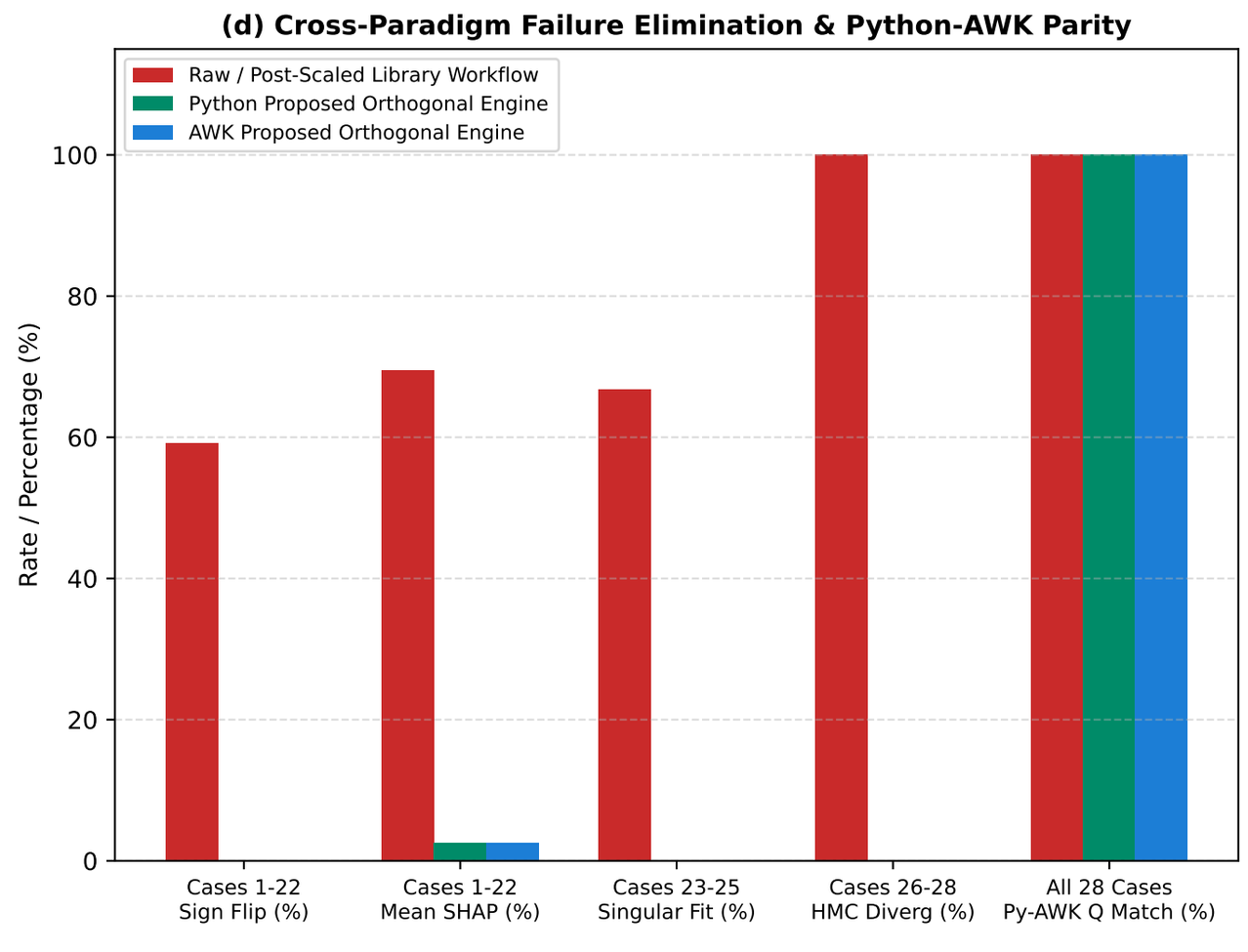

(a) All 28 Cohorts: Python vs. AWK Normalized Main-Effect Recovery
Normalized Main Effect $\hat{\beta}_1/\hat{\beta}_{\mathbb{Q},1}$
Medical Cohort Case ID (Cases 1-22: Freq/ML, 23-25: LMM, 26-28: Bayes HMC)
Exact Rational $\mathbb{Q}$ Baseline (+1.000)
Zero / Sign-Flip Boundary (0.0)
Python Proposed Orthogonal / Exact Q
AWK Proposed Orthogonal / Exact Q
Raw Uncentered Fit (Clipped to [-2.5, +2.5])


(b) Numerical Discrepancy Between Python and AWK Engines
$\log_{10}|\Delta\beta_1|$ (Floored at 10^{-16})
Medical Cohort Case ID (Cases 1-28)
Strict Verification Gate ($\varepsilon = 10^{-12}$)
$\log_{10}|\hat{\beta}^{\text{prop}}_{1,\text{Py}} - \hat{\beta}^{\text{prop}}_{1,\text{AWK}}|$ (Machine Floor 10^{-16})
$\log_{10}|\tilde{\beta}^{\text{raw}}_{1,\text{Py}} - \tilde{\beta}^{\text{raw}}_{1,\text{AWK}}|$ (Machine Floor 10^{-16})


(c) Unified Conditioning Metrics Across All 28 Cohorts (Python & AWK)
$\log_{10}$(Ill-Conditioning Factor)
Medical Cohort Case ID (Cases 1-28)
Raw Uncentered Ill-Conditioning (Python == AWK)
Proposed Orthogonal Conditioning (Python == AWK)
Cases 1-22
Log10(VIF)
C23-25
Log10(θ1/θ3)
C26-28
Log10(κ2)


(d) Cross-Paradigm Failure Elimination & Python-AWK Parity
Rate / Percentage (%)
Raw / Post-Scaled Library Workflow
Python Proposed Orthogonal Engine
AWK Proposed Orthogonal Engine
Cases 1-22
Sign Flip (%)
Cases 1-22
Mean SHAP (%)
Cases 23-25
Singular Fit (%)
Cases 26-28
HMC Diverg (%)
All 28 Cases
Py-AWK Q Match (%)